

From Election Forensics to Lottery Auditing: An Analysis of Korean Lotto 6/45

Minseong Kim

mkimacad@gmail.com

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Abstract

Statistical methods developed for election forensics - including digit-pattern tests, simulation-adjusted chi-square statistics, and summary-statistic variance tests - encode a general logic for detecting non-random structure in sequences of numbers that purport to be drawn uniformly at random. We argue that this logic transfers naturally to the auditing of state-run lotteries, where the fairness guarantee is both mathematically precise and publicly salient. We port, reformulate, and extend eleven tests from the election-forensics literature into a coherent lottery-audit framework, and validate the framework empirically against 1,204 consecutive draws of the Korean Lotto 6/45 (draws 1–1,204). The data are partitioned into five non-overlapping temporal windows to assess epoch-wise stability. The key empirical finding is that while isolated failures occur in five of 75 test-window combinations (6.7%), *no single test fails in more than one time window*. We show that this non-replication pattern is the appropriate fairness criterion in the presence of inter-test dependence, and that it is far more informative than a naive comparison of observed failures to a multiple-testing expectation computed under a false independence assumption. Beyond the empirical verdict, the paper’s primary contribution is the methodological toolkit itself: a reproducible, open-source audit pipeline that any researcher or regulator can apply to any 6/45-style lottery.

Keywords: lottery fairness audit; election forensics; digit tests; chi-square; temporal stability; Korean Lotto 6/45; multiple testing.

1 Introduction

State-run lotteries occupy an unusual position in public finance: they are monopoly revenue instruments whose legitimacy rests entirely on the credibility of a randomness

guarantee. Citizens who purchase tickets implicitly trust that every combination of six numbers is equally likely to be drawn, and that no information about future draws can be inferred from past ones. This guarantee is, in principle, mathematically precise. In practice, however, it depends on physical mechanisms - ball sets, pneumatic machines, draw procedures - that can drift, age, or be manipulated. Independent statistical auditing is therefore both scientifically tractable and socially important.

The election-forensics literature has, over the past two decades, developed a rich toolkit for testing whether reported vote counts are consistent with a fair counting process (Mebane, 2008; Shikano and Mack, 2011; Beber and Scacco, 2012). These methods share a common logic: they identify statistical signatures that a *genuinely random* process should exhibit - uniform digit distributions, predictable variances, absence of serial correlation - and test whether reported numbers deviate from those signatures. The inferential target differs from classical hypothesis testing: the analyst does not seek to estimate an unknown parameter but to assess whether the *generating process* is consistent with a specific null model.

We observe that this logic applies with equal, arguably greater, force to lottery auditing. The null model for a fair lottery is not merely plausible but precisely specified by combinatorics. Every marginal and joint distribution of draw statistics can be computed exactly or efficiently approximated. The audit problem is therefore a rare case where the statistical null is known, not assumed.

This paper makes three contributions. First, we systematically port eleven test statistics from the election-forensics literature into a lottery-audit framework, deriving the correct null distributions for the lottery setting (which differ non-trivially from the election context in several cases). Second, we introduce a *replication-logic* criterion for evaluating multi-test, multi-window audit results that does not require the independence of tests - a requirement that is plainly violated in any realistic test battery applied to the same dataset. Third, we validate the framework empirically on 1,204 consecutive draws of the Korean Lotto 6/45, the most widely played lottery in South Korea, providing a public benchmark for future audits.

The remainder of the paper is organized as follows. Section 2 reviews the Korean Lotto 6/45 and the election-forensics methods we adapt. Section 3 defines the eleven audit tests and their null distributions. Section 4 develops the replication-logic criterion. Section 5 describes the data. Section 6 presents the empirical results. Section 7 concludes.

2 Background

2.1 Korean Lotto 6/45

The Korean Lotto 6/45 (hereafter *Lotto*) is operated by the Korea Lottery Commission under the supervision of the Ministry of Economy and Finance. Each draw selects six balls uniformly without replacement from a pool of 45 numbered balls (1 through 45), plus one additional bonus ball. Draws are held every Saturday. The first draw was held in December 2002, and draw records are publicly accessible through the operator’s website.

Under the null hypothesis of a fair draw, the six winning numbers constitute a simple random sample of size $k = 6$ from the population $\{1, 2, \dots, N\}$ with $N = 45$. The total number of possible outcomes is $\binom{45}{6} = 8,145,060$.

2.2 Election Forensics: Relevant Methods

The election-forensics literature offers several classes of tests relevant to lottery auditing.

Digit tests. [Beber and Scacco \(2012\)](#) and [Mebane \(2008\)](#) exploit the observation that fabricated numbers tend to deviate from the digit distributions predicted by Benford’s law or uniform last-digit distributions. In the lottery context, the last-digit distribution is not uniform over $\{0, \dots, 9\}$ because the ball pool $\{1, \dots, 45\}$ contains unequal numbers of values ending in each digit; correcting for this structural non-uniformity is essential for a well-calibrated test.

Variance tests. [Shikano and Mack \(2011\)](#) propose comparing the observed variance of summary statistics (e.g., vote-share distributions) to the theoretical variance under a null model, arguing that fabricated data tends to exhibit suppressed variance. We adapt this idea to the draw-sum, odd-ball-count, and draw-range statistics of lottery draws.

Simulation-adjusted chi-square. [Shikano and Mack \(2011\)](#) also propose rescaling chi-square statistics by the ratio of the textbook critical value to the empirical 95th percentile from simulations under the null, correcting for distributional approximation errors. This is particularly valuable in the lottery context, where the exact null distribution of the ball-frequency chi-square is known to differ from the $\chi^2(44)$ asymptotic approximation at moderate sample sizes.

Mechanical-bias indicators. [Beber and Scacco \(2012\)](#) observe that round numbers ending in 0 or 5 tend to be over-reported in fabricated data. In the lottery context, this test functions as a mechanical-bias indicator: a physical ball machine with worn or unevenly weighted balls might exhibit systematic bias toward or away from balls whose numbers share a digit pattern.

3 The Eleven-Test Audit Framework

We organize the eleven tests into five conceptual domains: *frequency*, *digit pattern*, *draw-level distribution*, *dependence structure*, and *temporal stability*. Table 1 provides an overview.

Table 1: The eleven audit tests, organized by conceptual domain.

ID	Name	Null hypothesis tested	Source
<i>A. Frequency domain</i>			
L1	Ball-frequency χ^2	Each ball appears with equal marginal probability 6/45	Natural
L8	Simulation-adjusted χ^2	Same as L1, with finite-sample correction	Ported F7d
<i>B. Digit-pattern domain</i>			
L2	Last-digit (corrected null)	Last digits follow the non-uniform population distribution	Ported F2
L6	0s-and-5s bias	Balls ending in 0 or 5 appear at the population rate 9/45	Ported F8
L7	Last-digit mean & variance	Last-digit mean and variance match population moments	Ported F9
<i>C. Draw-level distribution domain</i>			
L3	Draw-sum distribution	Draw sums follow the exact hypergeometric convolution	Natural
L4	Odd/even and high/low splits	Split counts follow the hypergeometric distribution	Natural
L5	Pairwise co-occurrence	Each ball pair co-appears at rate $\binom{43}{4}/\binom{45}{6}$	Natural
<i>D. Dependence domain</i>			
L9	Temporal autocorrelation	Per-ball indicator series are serially uncorrelated	Natural
L11	Cross-ball correlation	Ball indicators have structural Pearson correlation $-1/44$	Natural
<i>E. Variance-stability domain</i>			
L10	Summary-statistic variance	Variance of sum, odd count, and range match theory	Ported F3

3.1 Frequency Domain (L1, L8)

L1 - Ball-frequency chi-square. Let c_b denote the count of draws in which ball $b \in \{1, \dots, 45\}$ appears, over n draws. Under the null, $\mathbb{E}[c_b] = nk/N = 6n/45$. The test statistic is

$$\chi_{L1}^2 = \sum_{b=1}^{45} \frac{(c_b - \hat{e}_b)^2}{\hat{e}_b}, \quad \hat{e}_b = \frac{6n}{45}, \quad (1)$$

with asymptotic null distribution $\chi^2(44)$.

L8 - Simulation-adjusted chi-square. The asymptotic $\chi^2(44)$ approximation can be inaccurate at moderate n . Following Shikano and Mack (2011), we draw $S = 500$ synthetic fair datasets of n draws, compute χ_{L1}^2 for each, and let $\hat{q}_{0.95}$ denote the empirical 95th percentile. The simulation-rescaled statistic is

$$\tilde{\chi}_{L8}^2 = \chi_{L1}^2 \cdot \frac{q_{0.95}^{\text{book}}}{\hat{q}_{0.95}}, \quad (2)$$

where $\chi_{44}^2(0.95)$ is the textbook critical value. The empirical p -value is $\Pr(\chi_{\text{sim}}^2 \geq \chi_{L1}^2)$.

3.2 Digit-Pattern Domain (L2, L6, L7)

Corrected null for last digits. A critical methodological point, often overlooked when porting election tests to the lottery context, is that the last digits of balls $\{1, \dots, 45\}$ are *not* uniformly distributed over $\{0, 1, \dots, 9\}$. The digit 0 appears only in balls 10, 20, 30, 40 (four balls), while digits 1–5 appear five times each (e.g., 1, 11, 21, 31, 41 for digit 1) and digits 6–9 appear four times each. Denoting the population proportion of last digit d as π_d , we have

$$\pi_d = \frac{|\{b \in \{1, \dots, 45\} : b \bmod 10 = d\}|}{45}. \quad (3)$$

L2 - Last-digit test. Let o_d denote observed appearances of balls with last digit d across all nk ball appearances. The expected count is $e_d = nk \pi_d$, rescaled to match $\sum o_d$. The statistic χ_{L2}^2 is computed against this corrected null and compared to $\chi^2(9)$.

L6 - 0s-and-5s test. Let $p_{05} = 9/45 = 1/5$ denote the population proportion of balls ending in 0 or 5 (balls 5, 10, 15, 20, 25, 30, 35, 40, 45). Let $\hat{p}$ be the observed proportion. The test statistic is

$$Z_{L6} = \frac{\hat{p} - p_{05}}{\sqrt{p_{05}(1 - p_{05})/(nk)}}, \quad (4)$$

with two-tailed p -value from the standard normal.

L7 - Last-digit mean and variance. The population mean and variance of last digits under the corrected null are

$$\mu_{\text{ld}} = \sum_{d=0}^9 d \pi_d, \quad (5)$$

$$\sigma_{\text{ld}}^2 = \sum_{d=0}^9 d^2 \pi_d - \mu_{\text{ld}}^2. \quad (6)$$

Mean is tested by a Z -test with $\text{se} = \sigma_{\text{ld}}/\sqrt{nk}$. Variance is tested by a two-tailed chi-square test with $nk - 1$ degrees of freedom.

3.3 Draw-Level Distribution Domain (L3, L4, L5)

L3 - Draw-sum distribution. The sum $S = \sum_{j=1}^6 b_j$ of a fair draw has a known exact distribution computable by dynamic programming over the combinatorial space. Its mean and variance are

$$\mathbb{E}[S] = k \frac{N+1}{2} = 6 \times 23 = 138, \quad (7)$$

$$\text{Var}[S] = k \frac{N^2-1}{12} \frac{N-k}{N-1}. \quad (8)$$

We apply (a) a Z -test on the sample mean of draw sums, and (b) a χ^2 goodness-of-fit test against the exact sum distribution after merging bins to ensure expected counts ≥ 5 .

L4 - Odd/even and high/low splits. Let X_i be the count of odd balls (or high balls, defined as those ≥ 23) in draw i . Under the null, $X_i \sim \text{Hypergeometric}(N, N_{\text{odd}}, k)$ where $N_{\text{odd}} = 23$. We test goodness-of-fit of the empirical distribution of $\{X_i\}$ against this hypergeometric null.

L5 - Pairwise co-occurrence. For each ordered pair (a, b) with $a < b$, let c_{ab} be the count of draws containing both a and b . Under the null, $\mathbb{E}[c_{ab}] = n \binom{43}{4} / \binom{45}{6}$. The test statistic is $\chi_{L5}^2 = \sum_{(a,b)} (c_{ab} - \hat{e})^2 / \hat{e}$ over all 990 pairs.

3.4 Dependence Domain (L9, L11)

L9 - Temporal autocorrelation. For ball b , define the indicator series $X_{b,i} = \mathbf{1}[b \in \text{draw}_i]$. We compute the sample autocorrelation $\hat{\rho}_b(\ell)$ at each lag $\ell = 1, \dots, \mathcal{L}$ for every ball, then aggregate via a Ljung–Box statistic

$$Q = n(n+2) \sum_{b=1}^{45} \sum_{\ell=1}^{\mathcal{L}} \frac{\hat{\rho}_b(\ell)^2}{n-\ell}, \quad (9)$$

compared to $\chi^2(45\mathcal{L})$. We use $\mathcal{L} = 100$ lags in the reported audits.

L11 - Cross-ball correlation. Under the null, the Pearson correlation between the indicator series of any two distinct balls is $-1/(N-1) = -1/44 \approx -0.0227$, a structural consequence of sampling without replacement (Hoeffding, 1948). We apply a Fisher Z -transformation to each of the 990 pair correlations and construct a global chi-square test over the Z -statistics.

3.5 Variance-Stability Domain (L10)

Following [Shikano and Mack \(2011\)](#)’s variance test (their F3), we test whether the observed variance of three draw-level summary statistics (draw sum S_i , odd-ball count X_i^{odd} , and draw range $R_i = \max_i - \min_i$) matches the theoretical variance under the null. For each statistic with theoretical variance σ_{theo}^2 , the test statistic is

$$\chi_{L10}^2 = \frac{(n-1)s^2}{\sigma_{\text{theo}}^2} \sim \chi^2(n-1), \quad (10)$$

where s^2 is the sample variance (with $n-1$ denominator). The theoretical variance of draw range is estimated by Monte Carlo (10,000 simulated draws) because no closed form is available.

4 The Replication-Logic Criterion

4.1 The Problem with Naive Failure Counting

A common, but flawed, approach to interpreting multi-test audit results is to compare the number of observed failures (tests with p -value < 0.05) to the number expected under the global null. Under complete independence, 75 tests at $\alpha = 0.05$ yield an expected $75 \times 0.05 = 3.75$ false positives. Observing 5 failures might then be interpreted as only a modest excess, consistent with chance.

This reasoning has two serious defects in the present setting.

Defect 1: Inter-test dependence. The eleven tests are not mutually independent. Tests L1 and L8 are derived from the same ball-count vector. Tests L2, L6, and L7 are all functions of the last digits of the same ball appearances. Tests L3 and the sum-variance subtest of L10 both use the draw-sum sequence. Tests L4 and the odd-count subtest of L10 both decompose draws by odd/even membership. Tests L9 and L11 both probe the dependence structure of the same ball-indicator matrix. The *effective* number of independent tests is substantially lower than 75 (or 15 per window), and no correction formula is available without strong distributional assumptions.

Defect 2: Within-window independence. Even if the 15 tests within each window were independent, the five windows are not independent experiments - they consist of consecutive draws from the same physical equipment. Any slow-drifting systematic bias would induce positive correlation in failures across windows, making multi-window failure co-occurrence less surprising than it would be under a fully independent design.

4.2 The Replication-Logic Criterion

We propose the following criterion for assessing fairness in a multi-test, multi-window audit.

Definition 1 (Replication consistency). *An audit result is replication-consistent with fairness if no single test fails in more than one non-overlapping time window.*

The rationale is as follows. A genuine, physically grounded violation of fairness - worn ball bearings, biased pneumatics, manipulated draw procedures - operates at the level of the generating mechanism, not the statistical test. Such a violation would produce a persistent signal that manifests repeatedly across time. A test that is sensitive enough to detect the violation in one window of n draws would, with high probability, also detect it in adjacent windows of comparable size, because the underlying cause has not been removed.

By contrast, a false positive generated by chance has probability approximately 0.05 per window per test, and consecutive false positives for the same test in independent windows occur with probability approximately $0.05^2 = 0.0025$. In a battery of 11 tests over 5 windows, the expected number of tests that fail in two or more windows is $11 \times \binom{5}{2} \times 0.05^2 \approx 0.28$ - that is, such “double failures” are rare under the null, making their absence a powerful fairness indicator.

Remark 1. *The replication-logic criterion does not require the independence of tests within a window. It operates across windows: a single failure in one window is neither surprising nor informative under H_0 ; a consistent failure in the same test across multiple windows constitutes evidence against H_0 regardless of what other tests show.*

Remark 2. *The criterion is conservative in the sense that inter-window positive correlation (arising from shared equipment) makes double failures more likely under both H_0 and H_1 than under independence. However, the increase in probability under H_1 is expected to be far larger than under H_0 , since a persistent physical cause amplifies correlation far beyond the level induced by shared equipment alone.*

5 Data

We use the complete publicly available draw history of the Korean Lotto 6/45, comprising draws 1 through 1,204. Draw records were obtained from the official Korea Lottery Commission website and contain, for each draw, the draw number, draw date, six main winning numbers, and one bonus number. Only the six main winning numbers are used in the audit; the bonus number is excluded.

The 1,204 draws span from December 2002 to approximately mid-2025. To assess temporal stability, we partition the draws into five non-overlapping, roughly equal-length windows:

Window	Draw range	n	Approx. years
W1	1–250	250	2002–2007
W2	251–500	250	2007–2012
W3	501–750	250	2012–2017
W4	751–1000	250	2017–2021
W5	1001–1204	204	2021–2025

The final window (W5) contains 204 draws because the total draw count is not a multiple of 250. Tests based on $\chi^2(n-1)$ statistics (L10) use the window-specific n .

No preprocessing beyond removing the bonus number is applied. Draws are sorted by draw number; no draws are missing within the range.

6 Results

6.1 Window-by-Window Test Results

Table 2 presents the p -values for all 75 test-window combinations. Cells marked **FAIL** indicate p -value < 0.05 .

Five tests fail at $\alpha = 0.05$ across the full audit: L6 in W4, L4 (odd/even) in W5, L9 in W3, L10 (sum variance) in W1, and L10 (range variance) in W5. The minimum p -value across all failures is 0.014 (L10 sum variance, W1).

6.2 Application of the Replication-Logic Criterion

Table 3 summarizes which tests fail in which windows.

The key finding is that *no single test fails in more than one non-overlapping window*. The five failures are dispersed across five distinct test-window cells, one per window (save W2, which has no failures), with no test appearing in the failure set more than once.

This pattern is precisely what the replication-logic criterion predicts under the global null of a fair lottery. The estimated probability that a specific test fails in two or more windows by chance, assuming independence across windows and tests at $\alpha = 0.05$, is at most $1 - (1 - 0.05)^5 \approx 0.226$ for any one test, but the probability that *any* of the five failures replicates is approximately $5 \times 0.05 = 0.25$ per additional window, making replication-free failures entirely expected under H_0 .

The windows and the tests are not independent, making replication *more* likely under H_1 than under independence. The complete absence of replication therefore strengthens the fairness conclusion beyond what the independence calculation alone suggests.

Table 2: p -values for all 11 tests across 5 temporal windows. **FAIL** denotes p -value < 0.05 ; bold entries are failures. Tests with sub-components (L4, L7, L10) show the relevant sub-test.

ID	Test (sub-test)	W1	W2	W3	W4	W5
<i>A. Frequency domain</i>						
L1	Ball frequency $\chi^2(44)$	.980	.847	.961	.719	.344
L8	Sim.-adj. χ^2	.934	.678	.910	.492	.148
<i>B. Digit-pattern domain</i>						
L2	Last digit $\chi^2(9)$	.814	.357	.383	.486	.772
L6	0s-and-5s Z	.651	.071	.606	.024	.932
L7	LD mean Z	.691	.554	.853	.697	.325
L7	LD variance χ^2	.610	.560	.261	.915	.921
<i>C. Draw-level distribution domain</i>						
L3	Draw sum χ^2	.473	.677	.651	.119	.996
L4	Odd/even split χ^2	.634	.874	.734	.471	.041
L4	High/low split χ^2	.099	.782	.834	.598	.613
L5	Pairwise co-occurrence χ^2	.957	.848	.996	.894	.345
<i>D. Dependence domain</i>						
L9	Temporal autocorr. (Ljung-Box)	.633	.308	.045	.880	.921
L11	Cross-ball correlation χ^2	.997	.996	.848	.999	.999
<i>E. Variance-stability domain</i>						
L10	Variance - draw sum	.014	.858	.858	.873	.691
L10	Variance - odd count	.346	.856	.466	.138	.771
L10	Variance - draw range	.126	.999	.390	.454	.032
<i>Failures per window</i>		1	0	1	1	2

6.3 Robustness of the Corrected Digit Null

A secondary result concerns the importance of applying the corrected last-digit null. The balls $\{1, \dots, 45\}$ have a non-uniform last-digit distribution: digits 1–5 each appear in five ball labels, while digits 0 and 6–9 each appear in four. Tests L2, L6, and L7 all account for this structural non-uniformity. Application of a naive uniform null (as used in some election-forensics ports to lottery data) would introduce systematic bias: under the corrected null, the digit-domain tests pass comfortably in all five windows, while under the uncorrected uniform null several windows would show spurious failures due to the structural over-representation of digits 1–5. This correction is thus not merely a technical nicety but has substantive implications for audit conclusions.

6.4 Test Behavior Under the Null

The per-window p -values exhibit behavior consistent with a calibrated test battery applied to fair data. The median p -value across all 75 test-window combinations is approximately 0.64, well above 0.50, which is expected when the null is true (by symmetry of the uniform

Table 3: Failure map across tests and windows. A filled circle ($\bullet$) indicates p -value < 0.05 ; an open circle ($\circ$) indicates p -value < 0.10 . No test fails in more than one window.

Test (sub-test)	W1	W2	W3	W4	W5
L4 - Odd/even					$\bullet$
L6 - 0s-and-5s		$\circ$		$\bullet$	
L9 - Autocorr.			$\bullet$		
L10 - Sum var.	$\bullet$				
L10 - Range var.					$\bullet$

distribution on $[0, 1]$). The p -values from the frequency and cross-ball tests (L1, L8, L11) are consistently very high, reflecting the relatively large sample sizes and the precision of the null model. The digit-domain tests (L2, L7) are well-calibrated and show no sign of systematic inflation or deflation across windows.

7 Conclusion

We have presented an eleven-test statistical audit framework for lottery fairness, constructed by porting and reformulating methods from the election-forensics literature. Three methodological contributions distinguish the framework from prior approaches.

First, several tests require non-trivial adaptation to the lottery context. The corrected last-digit null (L2, L7) accounts for the structural non-uniformity of digit frequencies in the ball pool $\{1, \dots, 45\}$. The temporal Ljung–Box statistic (L9) is aggregated across all 45 per-ball indicator series. The structural correlation null of $-1/44$ (L11) follows from the without-replacement sampling mechanism.

Second, the simulation-adjusted chi-square (L8), ported from [Shikano and Mack \(2011\)](#), provides a finite-sample correction to the ball-frequency test that is particularly valuable at the sample sizes (200–250 draws) typical of epoch-wise subgroup analysis.

Third, and most importantly, we introduce the *replication-logic criterion* as the appropriate fairness verdict in the presence of inter-test dependence: the audit result is replication-consistent with fairness if no single test fails in more than one non-overlapping temporal window. Applied to 1,204 draws of the Korean Lotto 6/45 partitioned into five windows, the criterion is satisfied: five isolated failures are observed, each in a distinct test-window cell, with no test failing more than once. This pattern is precisely what a well-calibrated audit of genuinely fair data should produce.

The framework is implemented as an open-source Python pipeline and can be applied to any 6/45-style lottery with a CSV draw history.

A Code availability

The code is available at https://github.com/mkimacad/mkimacad.github.io/tree/main/korean_lotto_forensics.

B Exact Null Distribution of the Draw Sum

The exact probability mass function of the draw sum $S = \sum_{j=1}^6 b_j$, where $\{b_1, \dots, b_6\}$ is a simple random sample without replacement from $\{1, \dots, 45\}$, is computed by dynamic programming. Let $f[j][s]$ denote the number of ways to choose j distinct elements from $\{1, \dots, 45\}$ that sum to s . The recurrence is

$$f[j][s] = f[j-1][s-b] \quad \text{accumulated over } b = 1, \dots, 45, \quad (11)$$

with boundary condition $f[0][0] = 1$ and $f[0][s] = 0$ for $s > 0$. The probability is $\Pr(S = s) = f[6][s] / \binom{45}{6}$. The support of S runs from $\sum_{i=1}^6 i = 21$ to $\sum_{i=40}^45 i = 255$.

C Structural Null Correlation for Ball Indicators

Let $X_b = \mathbf{1}[b \in \text{draw}]$ for $b \in \{1, \dots, 45\}$. Then $\mathbb{E}[X_b] = k/N = 6/45$, $\text{Var}(X_b) = (k/N)(1 - k/N)$. For $b \neq b'$,

$$\text{Cov}(X_b, X_{b'}) = \mathbb{E}[X_b X_{b'}] - \mathbb{E}[X_b] \mathbb{E}[X_{b'}] = \frac{k(k-1)}{N(N-1)} - \left(\frac{k}{N}\right)^2 = -\frac{k(N-k)}{N^2(N-1)}. \quad (12)$$

Dividing by $\text{Var}(X_b) = k(N-k)/N^2$ gives $\rho(X_b, X_{b'}) = -1/(N-1) = -1/44$. $\square$

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